

Comps Exams: Ecology Notes

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Ecology

I. Fundamentals of Ecology

1. Biology

Curtis, 1983 [pp. 49-75; 157-232]

Carbon chemistry

- $1s^2 2s^2 2p^2$
- $\Rightarrow sp^3$ hybrid orbitals; single covalent bonds
- $\Rightarrow sp^2$ hybrid orbitals; double covalent bonds
- $\Rightarrow sp^1$ hybrid orbitals; triple covalent bonds
- Oxidation states: -4 (CH_4) to +4 (CO_2)
- Reaction types: 1) Substitution, 2) Elimination, 3) Addition, 4) Rearrangement
- Functional groups: 1) -OH alcohols (hydroxy), 2) -COOH carboxylic acid (carboxyl), 3) -NH₂ amine (amino), 4) -CHO aldehyde, 5) -CHO- ketone, 6) -PO(OH)₂ phosphate

Carbohydrates (sugars)

- $(CH_2O)_n$ "hydrates of carbon"
- monosaccharides [glucose, $C_6H_{12}O_6$ (an aldose); fructose, $C_6H_{12}O_6$ (a ketose)]
- $\Rightarrow$ principal energy source: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O \Delta H = -673kcal$
- disaccharides [sucrose, condensation of glucose + fructose; 1 $\rightarrow$ 2 linkage]
- $\Rightarrow$ transport from photosynthesis
- polysaccharides: storage: starch (amylose, amylopectin) and glycogen; α linked glucose
- polysaccharides: structure: cellulose β linked glucose; chitin β linked glucose w/ N

Lipids (fats, oils, waxes)

- fats: glycerol + 3 fatty acids (esterification)
- $\Rightarrow$ 6x more energy stored (per wgt) than polysaccharides; hydrophobic
- $\Rightarrow$ organ cushion; insulation

- phospholipids: glycerol + 2 fatty acids (esterification) + 1 phosphate; hydrophobic and hydrophilic ends; cell membrane

Proteins (polypeptides)

- 20 amino acids: $H_2N - CHR - COOH$; only R varies
- amide (ester) linkage: $H_2N - CHR - CO - NH - CHR - COOH$; dipeptide
- primary (sequence), secondary (H bonds); tertiary (S cross links); quaternary (multiple polypeptides)
- structural: collagen (gelatin), keratin
- regulatory: insulin
- transport: hemoglobin
- storage: seed
- protective: antibodies
- toxins: diphtheria, botulism
- enzymes: sucrase

Nucleic acids

- purines (adenine, guanine), pyrimidines (cytosine, thymine) + sugar (ribose) + phosphate
- deoxyribonucleic acid, ribonucleic acid, tRNA, mRNA

Thermodynamics

- only $\sim 1\%$ of $1.3 \times 10^{24} cal$ of solar energy enters biosphere
- only $\sim 10\%$ energy transfer per trophic level
- Energetics of reactions: Gibbs free energy $\equiv \Delta G = \Delta H - T\Delta S$; exergonic $\equiv \Delta G \leq 0$; endergonic $\equiv \Delta G > 0$
- First Law: Total energy of an isolated system constant
- Second Law: Time's arrow; Transformations from potential energy sources accompanied by heat (friction; non-usable energy) or increase in entropy; Potential energy of final state LESS than initial state IF $\Delta G \leq 0$

Fundamental BioEnergy Cycle (Redox)

- Chloroplast: : $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ $\Delta G = +680kcal$
- Mitochondrion: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ $\Delta G = -680kcal$
- Metabolism: molecular construction rxns (anabolism) + molecular deconstruction rxns (catabolism)
- Metabolism: couple an endergonic rxn with hydrolysis of ATP (exergonic $\sim 7 kcal/mol$; often phosphorylation of a reactant(s))
Example: glucose + fructose + 2ATP $\rightarrow$ sucrose
- ATP: adenine + ribose + 3 phosphates = adenosine triphosphate

Enzymes

- Large globular proteins; catalysts; Reduction of activation potential
- Active site + substrate $\rightarrow$ products
- Coenzymes (vitamins, cations) needed often as electron acceptors (eg, nictotinamide adenine dinucleotide (NAD) = niacin)
- pH and temp affected; denatured = lost structure ($3^\circ, 4^\circ$)
- Enzymatic pathways prevent equilibrium; move rxns forward
- Auxotroph: lack critical enzyme fixing at some point in the pathway (missing or defective enzyme)
- Control of enzyme-mediated rxns with 1) activation enzymes; 2) temp/pH; 3) allosteric factors; 4) feedback inhibition; 5) competitive inhibition (sulfanilamide for PABA in folic acid synthesis); 6) noncompetitive inhibition (Pb poisoning binds to -SH and denatures enzymes); 7) irreversible inhibition (nerve toxins, penicillin on bacterial cell wall construction)

Glycolysis and Respiration

- Aerobic ATP Production: glycolysis + respiration $\equiv$ (Krebs cycle + electron transport)
- Glycolysis (in cytoplasm): Phosphorylation of glucose (by kinases) $\Rightarrow$ splits glucose to 2, 3C glyceraldehyde phosphates $\Rightarrow$ Oxidation to pyruvic acid $H_3CCOCOOH$; 9 steps
Glycolysis: $2 ATP + 2 NAD^+ \Rightarrow 4 ATP + 2NADH$
Glycolysis: controlled by negative feedback of ATP: phosphofructokinase is an allosteric enzyme with ATP itself being the allosteric factor

- Pyruvic acid can go aerobic (respiration) or anaerobic (fermentation to ethanol; oxygen debt to lactic acid)
- Respiration (in mitochondrion): pyruvic acid - $\text{CO}_2 + \text{CoenzymeA} \Rightarrow \text{acetyl CoA}$
Acetyl CoA enters Krebs cycle $\Rightarrow \text{ATP} + \text{NADH} + \text{CO}_2 + \text{H}_2\text{O}$
NADH enters oxidative phosphorylation (electron transport) $\Rightarrow \text{ATP} + \text{H}_2\text{O}$
- Paul Mitchell (Nobel chemistry, 1976) proposed radical (ingenious) idea that oxidative phosphorylation of ADP to ATP in mitochondria powered by proton gradient (like active transport); proton gradient creates pH gradient and voltage gradient; met with skepticism for 20 yrs

2. Ecology

Curtis, 1983 [pp. 951-993]

Weier, 1982 [pp. 347-371]

Odum, 1969

Hairston et al., 1960

Cardinale et al., 2012

Estes et al., 2011

J. J. Wiens et al., 2010

3. Evolution

Curtis, 1983 [pp. 928-945]

Ehrlich and Raven, 1969

Alleaume-Benharira et al., 2006

Jay Gould and Eldredge, 1993

Kauffman, 1993 [pp. 3-120; 173-286]

Beinhocker, 2007 [pp. 187-217]

Feng et al., 2010

II. Foundations of Landscape Ecology

1. Historical Development

Solnetsev, 1948

Troll, 1950

Watt, 1947

J. A. Wiens, 2008

2. Ecological Landscapes: Scales and Patterns

Turner et al., 1989

Wu, 2004

Legendre and Fortin, 1989

J. A. Wiens, 1989

3. Ecological Landscapes: Dynamics and Processes

Turner, 2010

Turner, 1989

Peterjohn and Correll, 1984

- exchangeable NH_4 , total Kjeldahl-N, dissolved NO_3 and NO_2 , exchangeable orthophosphate, total-P, millipore filtered vs. unfiltered samples, point centered quarter sampling technique Mitchell, 2010, throughfall,
- nutrient dynamics in ag (corn field) watershed upstream of broad leaved deciduous trees, Maryland; compare pathways of nutrient flux within and between habitats
- examine N,C,P transformations; samples from surface and shallow subsurface waters
- 1680 kg/ha limestone (April); 2-10-13 fertilizer (April); pre-emergent herbicide (May); liquid N (May); urea (July)
- 105 kg/ha N (63% organic, 22% NH_4^+ , 15% NO_3^-) PLUS 14 kg/ha N (46% organic, 20% NH_4^+ , 34% NO_3^-)
- spring input 13.9 kg/ha C (bulk precip) BUT spring watershed discharge 153 kg/ha C (death of soil biota) ==> every other season has C input proportional to rain but substantially less C discharge (eg, fall C bulk precip 5.7 kg/ha; watershed discharge C 0.42 kg/ha)
- pulse of C in Spring because killed soil biota !!
- Corn: nutrient uptake ≈ 250 kg N/ha/yr and ≈ 45 kg P/ha/yr; NPP $\approx 22,000$ kg C/ha/yr
- 28% of N and 23% P exported by corn harvest
- N retention poor in disturbed ecosystems ($\approx 10\%$); N retention excellent in undisturbed forest ecosystems ($\approx 90\%$)
- P retention average in disturbed ecosystems ($\approx 40\%$); P retention very good in undisturbed forest ecosystems ($\approx 80\%$)

- groundwater (subsurface flow) the major pathway of N loss from both crop field and forest
- forest throughfall N $\Rightarrow \approx 70\%$ retention by canopy
- forest throughfall P $\Rightarrow \approx 80\%$ supplemented by canopy
- Big question: What caused the decrease in groundwater NO₃ concentration in the riparian forest? Samples pulled below rhizosphere (at 1 m). Peterjohn and Correl hypothesize only vegetative uptake and denitrification. But incremental growth accounts for only 33%! Authors therefore default to denitrification. But bacteria now known to uptake nitrate and nitrite (nitrate and nitrite reductase enzymes)
- Riparian forest removed 4.1 Mg/ha particulates, 11 kg/ha particulate organic-N, 0.83 kg/ha NH₄, 2.7 kg/ha NO₃, 3 kg/ha particulate P from surface runoff
- Riparian forest removed 45 kg/ha NO₃ from subsurface flow

Wegner and Merriam, 1979

Walker et al., 2010

J. A. Wiens, 1976

4. Measuring and Modeling Ecological Landscapes

Dale and Fortin, 2014 (TBD)

Naveh, 1987

Lefkovitch and Fahrig, 1985

Scheller and Mladenoff, 2007

Scheller, Robert M., 2013

Scheller et al., 2007

III. Agroecology

1. Concepts and Rationales

Tilman et al., 2002

Godfray et al., 2010

Gliessman, 2015 [pp. 3-34; 301-344]

Wolfe, 2000

Beus and Dunlap, 1990

Rosset, 2000

Thompson, 2010

2. Biotics and Abiotics

Schmidt et al., 2011

Gliessman, 2015 [pp. 35-142]

3. Agroecological Interactions

Gliessman, 2015 [pp. 143-298]

Gianinazzi and Schuepp, 1994

Schnitzer et al., 2011

Smith et al., 2009

van der Putten et al., 2013

4. Milpa Culture

Jiménez-Osornio et al., 1999

Toledo et al., 2008

Benjamin et al., 2001

Dahlin et al., 2005

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